Beta Distribution

1 (A) Beta Distribution of first Kind

A continuous RV X is said to have a Beta distribution of first kind of its PDF is,

$$f_X(x) = \begin{cases} \frac{x^{m-1}(1-x)^{n-1}}{B(m,n)}, & x > 0 \\ 0, & \text{otherwise} \end{cases}$$

$$\left[\int_0^1 x^{m-1}(1-x)^{n-1}dx = B(m,n) \Rightarrow \int_0^1 \frac{x^{m-1}(1-x)^{n-1}}{B(m,n)}dx = 1\right]$$

2 Moments

$$\begin{aligned} \mu_r' &= E\left(X^r\right) \\ &= \int_0^1 x^r \frac{x^{m-1}(1-x)^{n-1}}{B(m,n)}dx \\ &= \frac{B(m+r,n)}{B(m,n)} = \frac{\Gamma(m+r)\Gamma n}{\Gamma(m+r+n)}\frac{\Gamma(m+n)}{\Gamma m\Gamma n} \\ &= \frac{\Gamma(m+r)\Gamma(m+n)}{\Gamma m\Gamma(m+r+n)} \end{aligned}$$

$$E(X) = \mu_1' = \frac{\Gamma(m+1)\Gamma(m+n)}{\Gamma m\Gamma(m+n+1)} = \frac{m}{m+n}$$

$$\begin{aligned} V(X) = \mu_2' - \mu_1'^2 &= \frac{\Gamma(m+2)\Gamma(m+n)}{\Gamma m\Gamma(m+n+2)} - \left(\frac{m}{m+n}\right)^2 \\ &= \frac{(m+1)m}{(m+n)(m+n+1)!} - \frac{m^2}{(m+n)^2} \\ &= \frac{mn}{(m+n)^2(m+n+1)} \end{aligned}$$

HM : $H = \frac{1}{\mu_1'} = \frac{1}{\frac{m+n-1}{m-1}} = \frac{m-1}{m+n-1}$ if $-1 > +m$

GM:

$$\begin{aligned} \log_e G &= E\left(\log_e X\right) \\ &= \int_0^1 \log X \frac{x^{m-1}(1-x)^{n-1}}{B(m,n)}dx \end{aligned}$$

Now, $\int_0^1 x^{m-1}(1-x)^{n-1}dx = B(m,n)$

$$\Rightarrow \frac{\partial}{\partial m}\int_0^1 x^{m-1}(1-x)^{n-1}dx = \frac{\partial}{\partial m}B(m,n)$$

$$\Rightarrow \int_0^1 \log_e x x^{m-1}(1-x)^{n-1}dx = \frac{\partial}{\partial m}B(m,n).$$

$$\Rightarrow \log_e G = \frac{\frac{\partial}{\partial m}B(m,n)}{B(m,n)} = \frac{\partial}{\partial m}\{\ln B(m,n)\}\frac{\partial}{\partial m}\left\{\log\frac{\sqrt{m+\sqrt{n}}}{\sqrt{m+n}}\right\}.$$

1. If $X \sim$ Beta (m, n) [Beta Distribution of first kind with parameter $m, 1$], find the distribution of $(1 - X)$. Hence, show that $B(n, n)$ is symmetric and compute the mean and median.

$$Y = 1 - X$$

$$F_Y(y) = P[Y \leqslant y] = P(1 - X \leqslant y) = P(X \geqslant 1 - y)$$

$$\frac{d}{dy}F_Y(y) = f_Y(y) = -f_X(1-y)(-1) = f_X(1-y)$$

$$= \begin{cases} \dfrac{(1-y)^{m-1}(1-(1-y))^{n-1}}{\beta(m,n)} & 0 < 1-y < 1 \\ 0 & \text{otherwise} \end{cases}$$

$$= \begin{cases} \dfrac{y^{n-1}(1-y)^{m-1}}{B(n,m)} & 0 < y < 1 \\ 0 & \text{otherwise} \end{cases}$$

$\therefore Y \sim B(n, m)$

$$f_X(x) = \frac{x^{n-1}(1-x)^{n-1}}{B(m,n)} = f_X(1-x), \quad 0 < x < 1$$

Clearly, the distribution is symmetric about $x = \frac{1}{2}$.

$\Rightarrow$ mean = median $= \dfrac{1}{2}$(if $n > 1$). If $0 < n < 1$, mode does not exist.

3 Mode

The value of x for which the PDF $f_X(x)$ is maximum.

$$\ln f_X(x) = (m-1)\ln x + (n-1)\ln(1-x) - \underbrace{\ln(B(m,n))}_{\text{constant}}$$

$$\frac{d}{dx}\left[\ln f_X(x)\right] = (m-1)\frac{1}{x} + (x-1)\frac{1}{(1-x)}(-1)$$

$$= \frac{m-1}{x} - \frac{n-1}{(1-x)}$$

$$= \frac{m-1-mx+x-nx+x}{x(1-x)}$$

$$= \frac{(m-1)-(m+n-2)}{x(1-x)}x$$

$\ln f_x(x)$ or $f_x(x)$ increases and then attains its maxima at $x = \frac{m-1}{m+n-2}$

$$\therefore \text{ Mode } = \frac{m-1}{m+n-2}.$$

2. Prove that $\Gamma\frac{1}{2}\Gamma 2n = 2^{2n-1}\Gamma n\Gamma n + \frac{1}{2}$. [Legendre's Duplication formula]

$$B(n,n) = \int_0^1 x^{n-1}(1-x)^{n-1}dx = 2\int_0^{1/2}\left(x - x^2\right)^{n-1}dx - (1)$$

$$y = 4\left(x - x^2\right)$$

$$\Rightarrow 4x - 4x^2 - y = 0$$

$$\Rightarrow 4x^2 - 4x + y = 0$$

$$\Rightarrow x^2 - x + \frac{y}{4} = 0 \Rightarrow x = \frac{1 \pm \sqrt{1 - 4 \cdot \frac{y}{4}}}{2}$$

$\frac{1+\sqrt{1-y}}{2}$ is not a feasible solution.

$$\therefore x = -\frac{1-\sqrt{1-y}}{2} \Rightarrow dx = -\frac{1}{2}\frac{1}{2\sqrt{1-y}}(-1)dy$$

$$\text{From (1), } 2\int_0^1 \left(\frac{y}{4}\right)^{n-1}\frac{1}{4}\frac{1}{\sqrt{1-y}}dy = \frac{1}{2^{2n-1}}\int_0^1 y^{n-1}(1-y)^{1/2-1}dy$$

$$B(n,n) = \frac{1}{2^{2n-1}}B\left(n,\frac{1}{2}\right)$$

$$\Rightarrow \frac{\Gamma n\Gamma n}{\Gamma 2n} = \frac{1}{2^{2n-1}}\frac{\pi\Gamma 1/2}{\Gamma(n+1/2)}$$

$$\Rightarrow \Gamma n\Gamma(n+\frac{1}{2}) = \frac{\sqrt{\pi}}{2^{2n-1}}\Gamma(2n)$$

$$\therefore \Gamma(\frac{1}{2})\Gamma(2n) = 2^{2n-1}\Gamma n\Gamma(n+\frac{1}{2}) \text{ [Proved].}$$

4 CDF

$$F_X(x) = P(X \le x) = \int_0^x \frac{t^{m-1}(1-t)^{n-1}}{B(m,n)}dt = \frac{\int_0^x t^{m-1}(1-t)^{n-1}dt}{\int_0^1 t^{m-1}(1-t)^{n-1}dt} = I_x(m,n)$$

which is known as Incomplete Beta function. Exact evaluation of $I_x(m,n)$ is not possible. The values of $I_x(m,n)$ for different (m,n) and it has been extensively tabulated by Karl Pearson which helps us to calculate the cumulative probabilities for Binomial and Negative Binomial distribution.

5 (B) Beta distribution of second Kind

A continuous random variable x is said to have a Beta distribution if its PDF is given by,

$$f_X(x) = \begin{cases} \dfrac{x^{m-1}}{B(m,n)(1+x)^{m+n}} & ,0 < x < \infty \\ 0 & \text{otherwise} \end{cases}$$

1. If $X \sim \text{Beta}_2(m,n)$, $Y \sim \frac{X}{1+X} \backsim \text{Beta}_1(m,n)$.

$$F_Y(y) = P(Y \leqslant y) = P\left(\frac{X}{1+X} \leqslant y\right) = P\left(X \leqslant \frac{y}{1-y}\right) = F_X\left(\frac{y}{1-y}\right)$$

As $0 < x < \infty \Rightarrow 0 < y = \frac{X}{1+X} < 1$

$$f_Y(y) = \begin{cases} f_X\left(\dfrac{y}{1-y}\right)\dfrac{(1-y)1+y}{(1-y)^2} & = f_X\left(\frac{y}{1-y}\right)\frac{1}{(1-y)^2}, 0 < y < 1. \\ 0 & \text{otherwise} \end{cases}$$

$$= \frac{1}{B(m,n)}\frac{\left(\frac{y}{1-y}\right)^{m-1}}{\left(1+\frac{y}{1-y}\right)^{m+n}\frac{1}{(1-y)^2}}, 0 < y < 1$$

$$= \frac{1}{B(m,n)}y^{m-1}(1-y)^{n-1}$$

$$Y = \frac{X}{1+X} \backsim B_1(m,n)$$

2. If $X \sim B_1(m,n), Y = \frac{X}{1-X} \sim B_2(m,n)$

$F_Y(y) = P(Y \leqslant y) = P\left(\frac{X}{1-X} \leqslant y\right) = P\left(X \leqslant \frac{y}{1+y}\right) = F_X\left(\frac{y}{1+y}\right)$

As $0 < x < 1 \Rightarrow 0 < y = \frac{x}{1-x} < \infty$.

$$f_Y(y) = \begin{cases} f_X\left(\frac{y}{1+y}\right)\frac{(1+y)-y1}{(1+y)^2} = f_X\left(\frac{y}{1+y}\right)\frac{1}{(1+y)^2}, & 0 < y < \infty. \\ 0 & \text{otherwise.} \end{cases}$$

$$= \begin{cases} \frac{1}{B(m,n)} & \left(\frac{y}{1+y}\right)^{m-1}\left(1-\frac{y}{1+y}\right)^{n-1}\frac{1}{(1+y)^2}, \quad 0 < y < \infty. \\ 0 & \text{otherwise.} \end{cases}$$

$$= \begin{cases} \frac{1}{B(m,n)}\frac{y^{m-1}}{(1+y)^{m-1}}\frac{(1+y-y)^{n-1}}{(1+y)^{n-1}}\frac{1}{(1+y)^2}, & 0 < y < \infty. \\ 0, & \text{otherwise} \end{cases} = \begin{cases} \frac{1}{B(m,n)}\frac{y^{m-1}}{(1+y)^{m+n}} \\ 0 \qquad \text{otherwise} \end{cases}$$

$Y = \frac{X}{1-X} \cap B_2(m,n)$.

3. Let, $f(x) = k\left(1+\frac{x^2}{n}\right)^{-\left(\frac{n+1}{2}\right)}, \quad x \in R$, be the PDF of X. Find k.

$$\Rightarrow 1 = \int_{-\infty}^{\infty} f_X(x)dx \Rightarrow k\int_{-\infty}^{\infty}\frac{1}{\left(1+\frac{x^2}{n}\right)^{\frac{n+1}{2}}}dx = 1, \quad t = \frac{x^2}{n}$$

$$\Rightarrow 2k\int_0^{\infty}\frac{1}{\left(1+\frac{x^2}{n}\right)^{\frac{n+1}{2}}}dx = 1 \qquad \Rightarrow dx = \sqrt{n}\frac{1}{2}\frac{1}{\sqrt{t}}dt$$

$$\Rightarrow 2\int_0^{\infty}\frac{1}{(1+t)^{\frac{n+1}{2}}}\frac{\sqrt{n}}{2}t^{\frac{1}{2}-1}dt = 1$$

$$\Rightarrow t\sqrt{n}\int_0^{\infty}\frac{t^{\frac{1}{2}-1}}{(1+t)^{\frac{n+1}{2}}}dt = 1$$

$$\Rightarrow k\sqrt{n}B\left(\frac{1}{2},\frac{n}{2}\right) = 1$$

$$\Rightarrow k = \frac{\sqrt{\frac{n+1}{2}}}{\frac{\sqrt{n}}{2}\sqrt{n}\sqrt{1/2}}$$

6 Moments

$$\begin{aligned} \mu_r' = E\left(X^r\right) &= \int_0^{\infty} x^r\frac{x^{m-1}}{B(m,n)(1+x)^{m+n}}dx \\ &= \frac{1}{B(m,n)}\int_0^{\infty}\frac{x^{m+r-1}}{(1+x)^{(m+n)+(n-r)}}dx \\ &= \frac{1}{B(m,n)}B(m+r,n-r) \\ &= \frac{\Gamma(m+n)}{\Gamma m\Gamma n}\frac{\Gamma(m+r)\Gamma(n-r)}{\Gamma(m+n)} \\ &= \frac{\Gamma(m+r)\Gamma(n-r)}{\Gamma m\Gamma n}; \quad m+r > 0 \Rightarrow r > -m; n-r > 0 \Rightarrow n > r \Rightarrow m < r < n \end{aligned}$$

$$\mu_1' = \frac{\Gamma(m+1)\Gamma(n-1)}{\Gamma m \Gamma n} = \frac{m!(n-2)!}{(m-1)!(n-1)!} = \frac{m}{(n-1)}$$

$$\therefore E(X) = \frac{m}{n-1}$$

$$\mu_2' = \frac{\Gamma(m+2)\Gamma(n-2)}{\Gamma m \Gamma n} = \frac{(m+1)!(n-3)!}{(m-1)!(n-1)!} = \frac{(m+1)m}{(n-1)(n-2)}$$

$$V(X) = \mu_2' - \mu_1'^2 = \frac{(m+1)m}{(n-1)(n-2)} - \frac{m^2}{(n-1)^2}$$

$$= \frac{m}{(n-1)}\left[\frac{m+1}{n-2} - \frac{m}{n-1}\right]$$

$$= \frac{m}{(n-1)^2(n-2)}$$